

GAIRA V7 — Phase 02.5

Latent Geometry of Spectral Motif Space

Figure report — all 25 figures with captions

Status	COMPLETE — analysis only; nothing refitted, no themes created
Frozen atlas	09ed804a40836f4a05a91ba10900cded
Frozen LSM registry	208482d6f7178b5b8f16cace91be55b0
Frozen CSM dictionary	0b4aa550ccefed3edabdbde5bae11c8d
Primary spectral metric	wasserstein
Primary fused geometry	weighted_similarity
Neighbourhoods computed on	D_primary_metric
Metric vs fused kNN agreement	0.508
Geometry verdict	mixed — overlapping continua
kNN chemical coherence	0.240 (p = 0.0000)
Graph modularity z	186.6 vs degree-preserving null
Provisional Phase 03 priors	10
Source-untestable motifs	8 of 50
Completed	2026-08-06 14:50:56 UTC

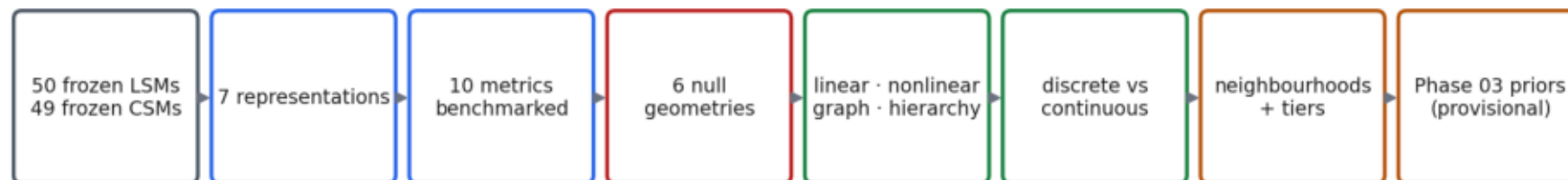
Committed source of record: reports/PHASE_02_5_LATENT_GEOMETRY_REPORT.md
Figures are embedded from the committed PNGs, so this PDF cannot drift from the run that produced them.

Figures

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Figure 01

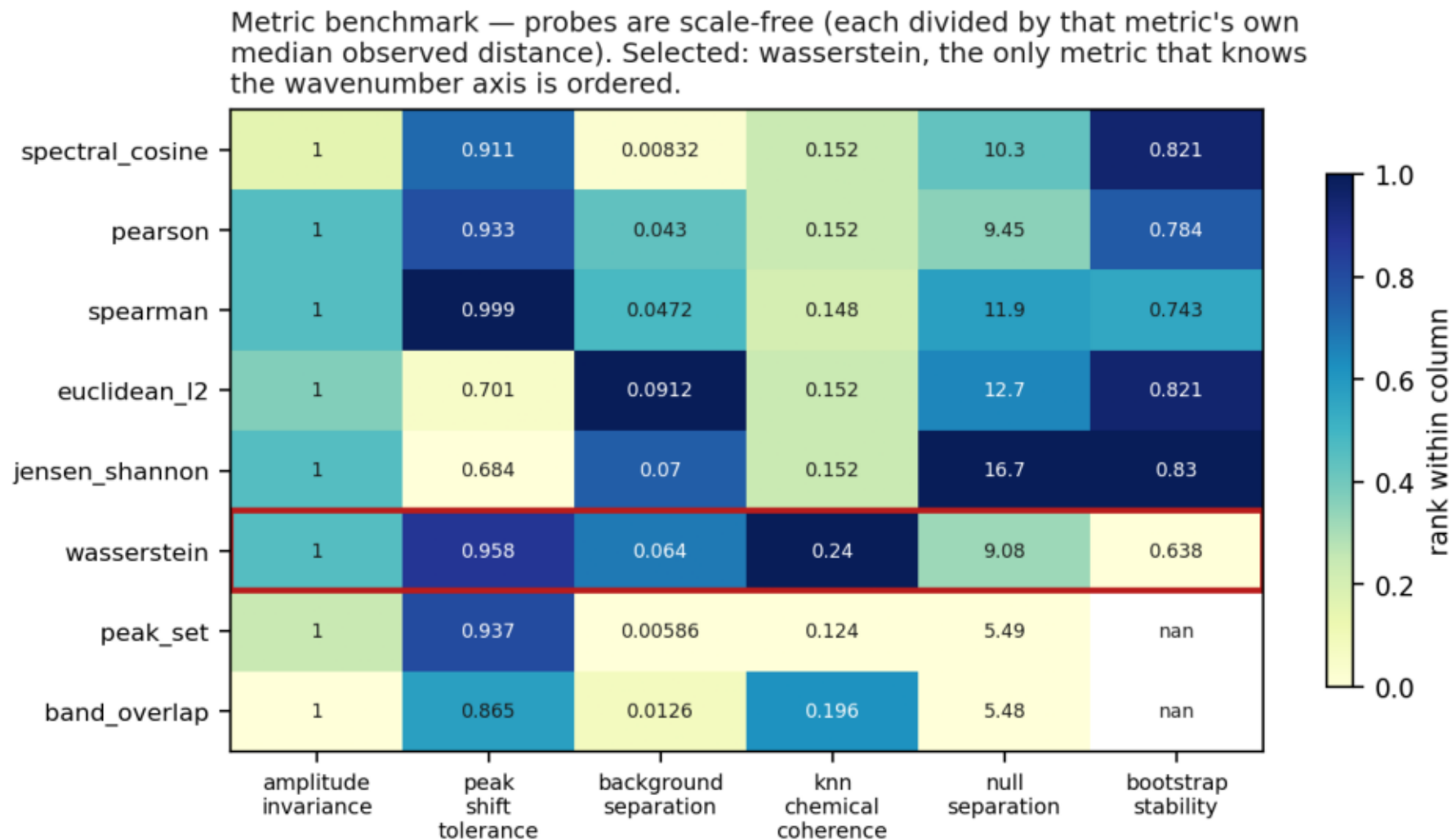
Phase 02.5 — latent geometry of spectral motif space



chemistry and source labels are excluded from every representation and every distance — revealed only at the neighbourhood step

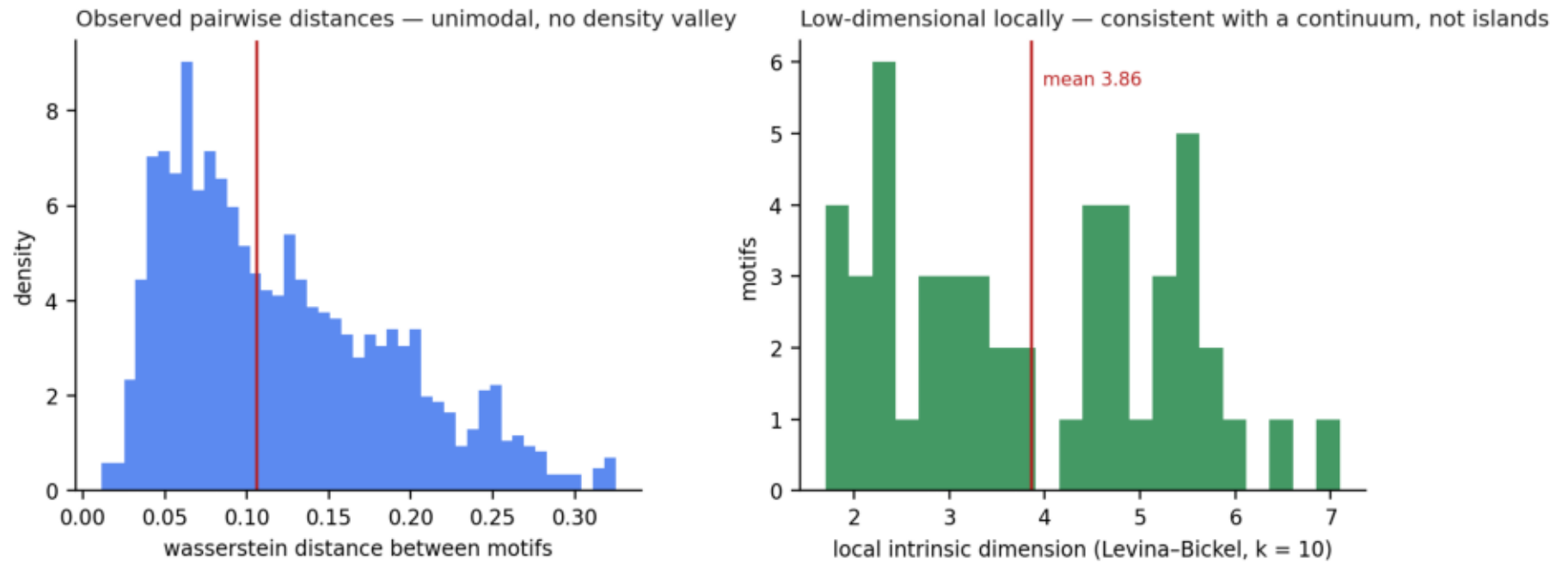
no dictionary is refitted · no themes are created

Figure 02



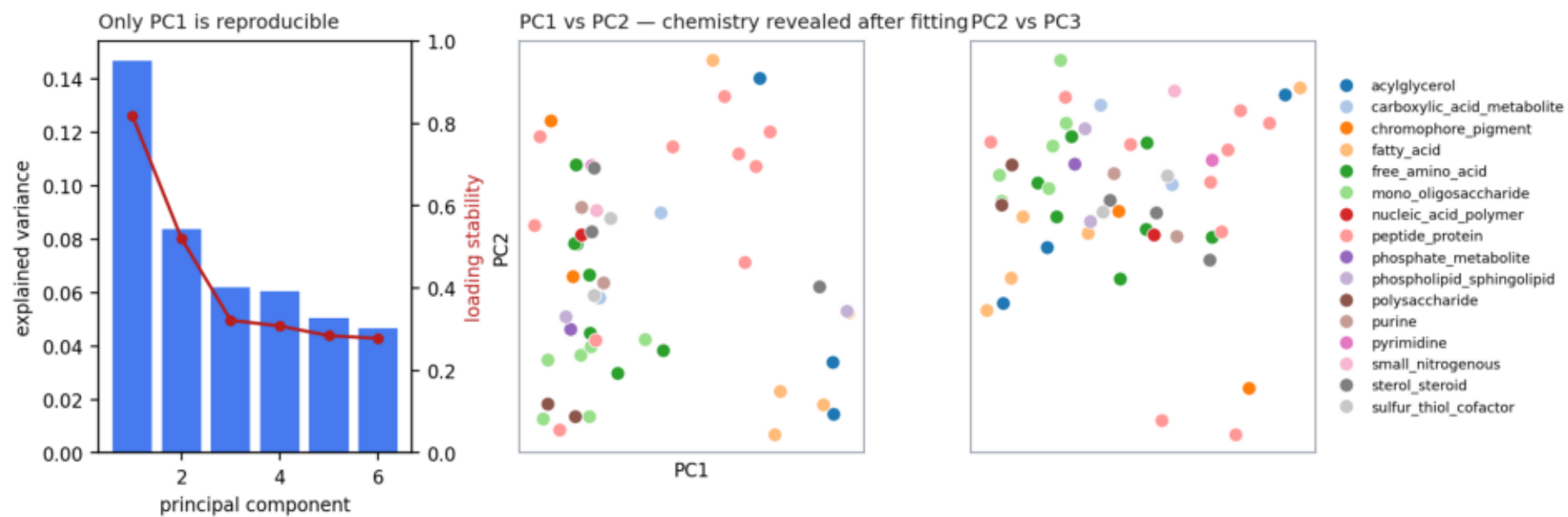
Metric benchmark. Probes are scale-free — each divided by that metric's own median observed distance — because a background separation of 0.106 under Euclidean and 0.006 under cosine cannot be compared until both are read against the scale each metric works on. Wasserstein selected.

Figure 03



Observed pairwise distances are unimodal with no density valley (depth 0.003), and local intrinsic dimension averages 3.86 against an ambient 676. Both point to a continuum rather than discrete islands.

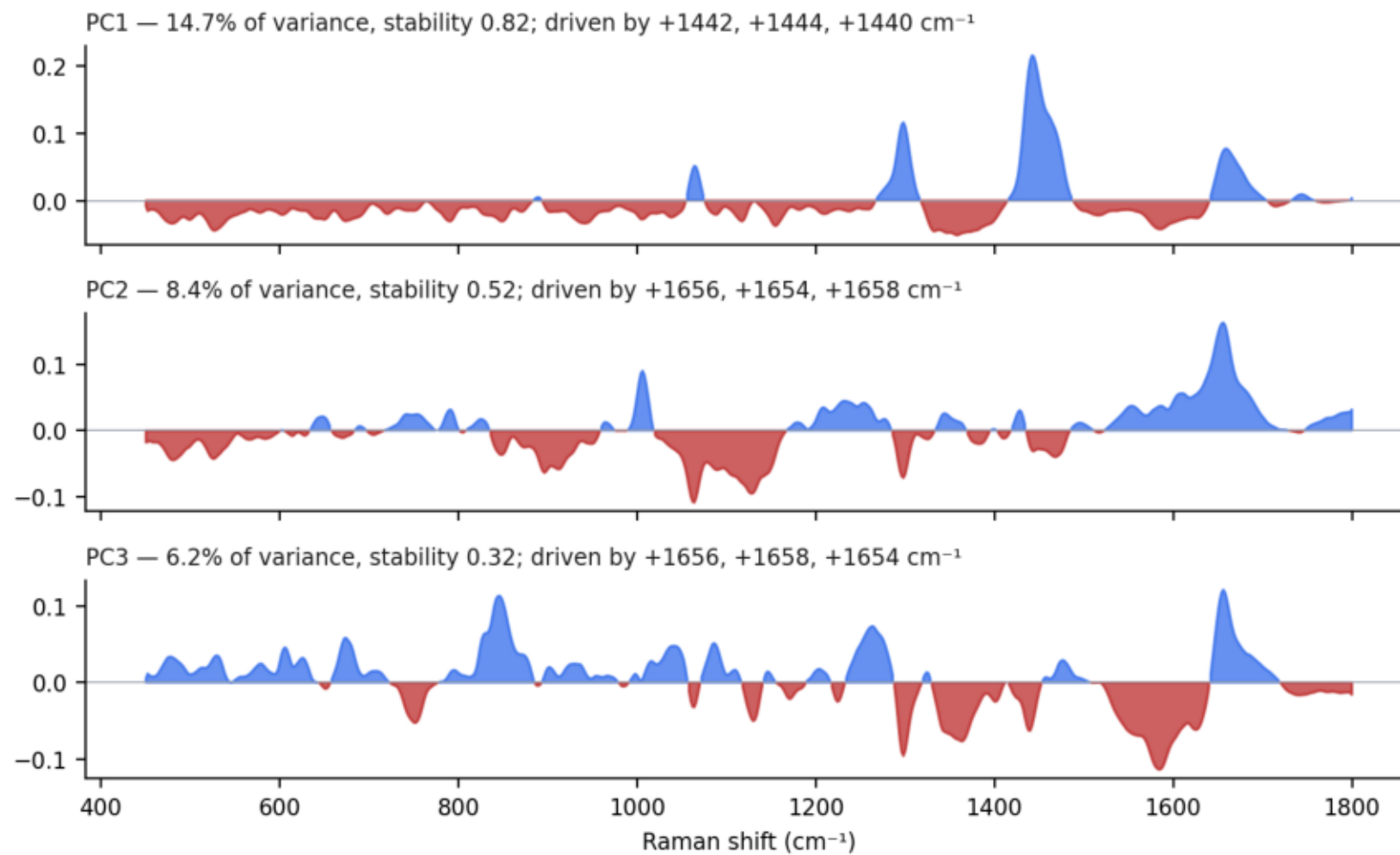
Figure 04



PCA scores. Only PC1 has a reproducible loading (stability 0.82); PC2 falls to 0.52 and PC3 to 0.32. Chemistry colour was attached after fitting.

Figure 05

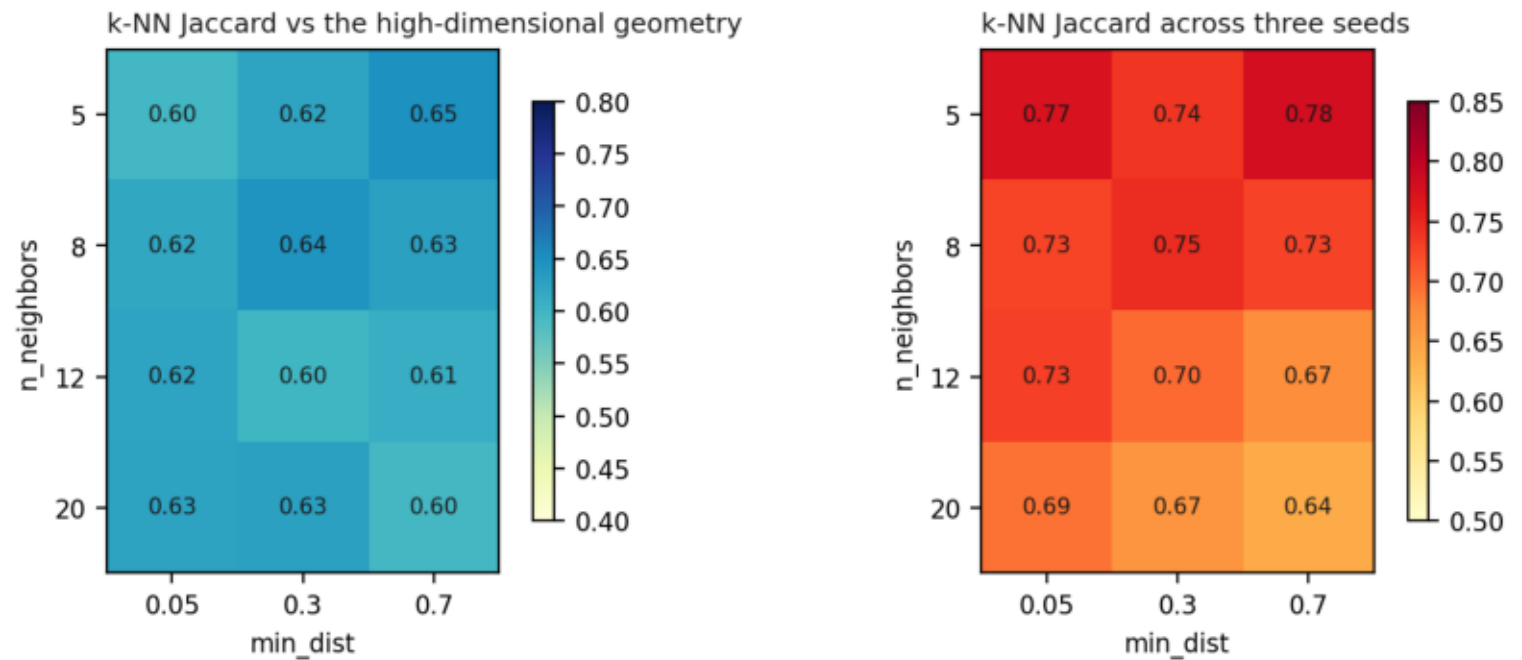
PCA loadings — what each axis is made of, spectroscopically



PCA loadings. PC1 is CH_2/CH_3 scissoring at 1442 cm^{-1} — an aliphatic-content axis, and the only component worth interpreting.

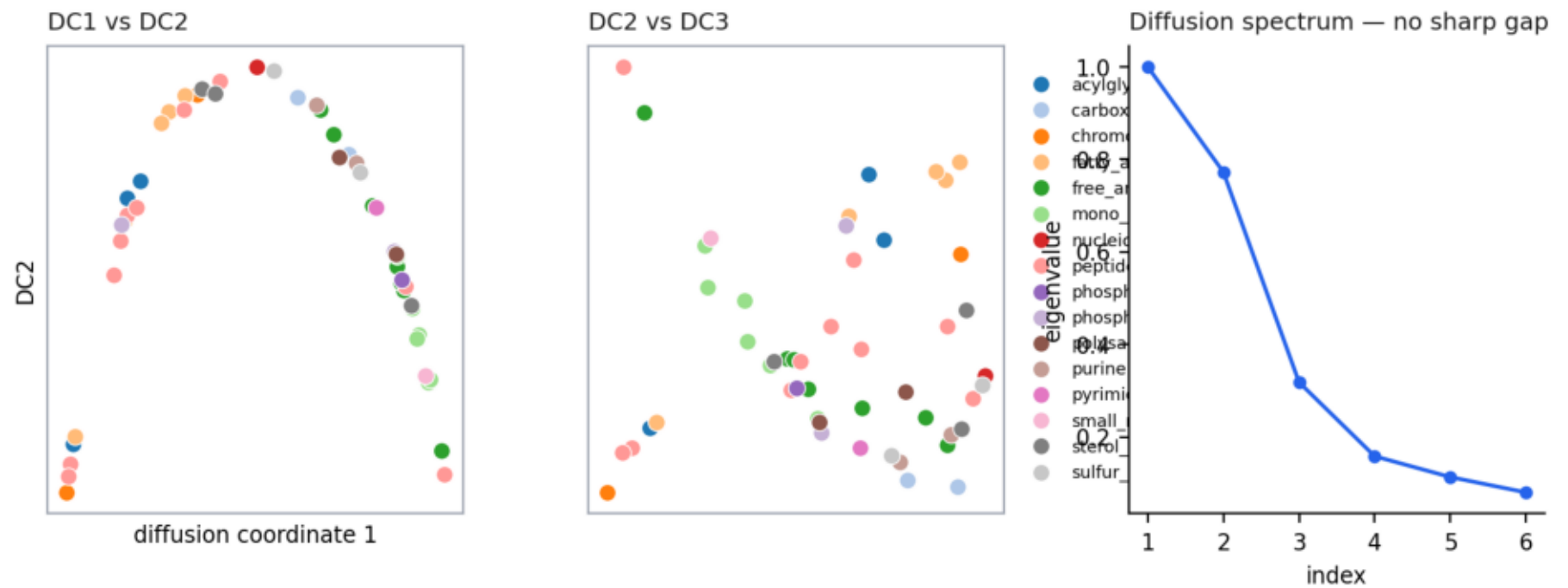
Figure 06

UMAP stability sweep — one layout is never evidence; the question is whether neighbours survive the parameters



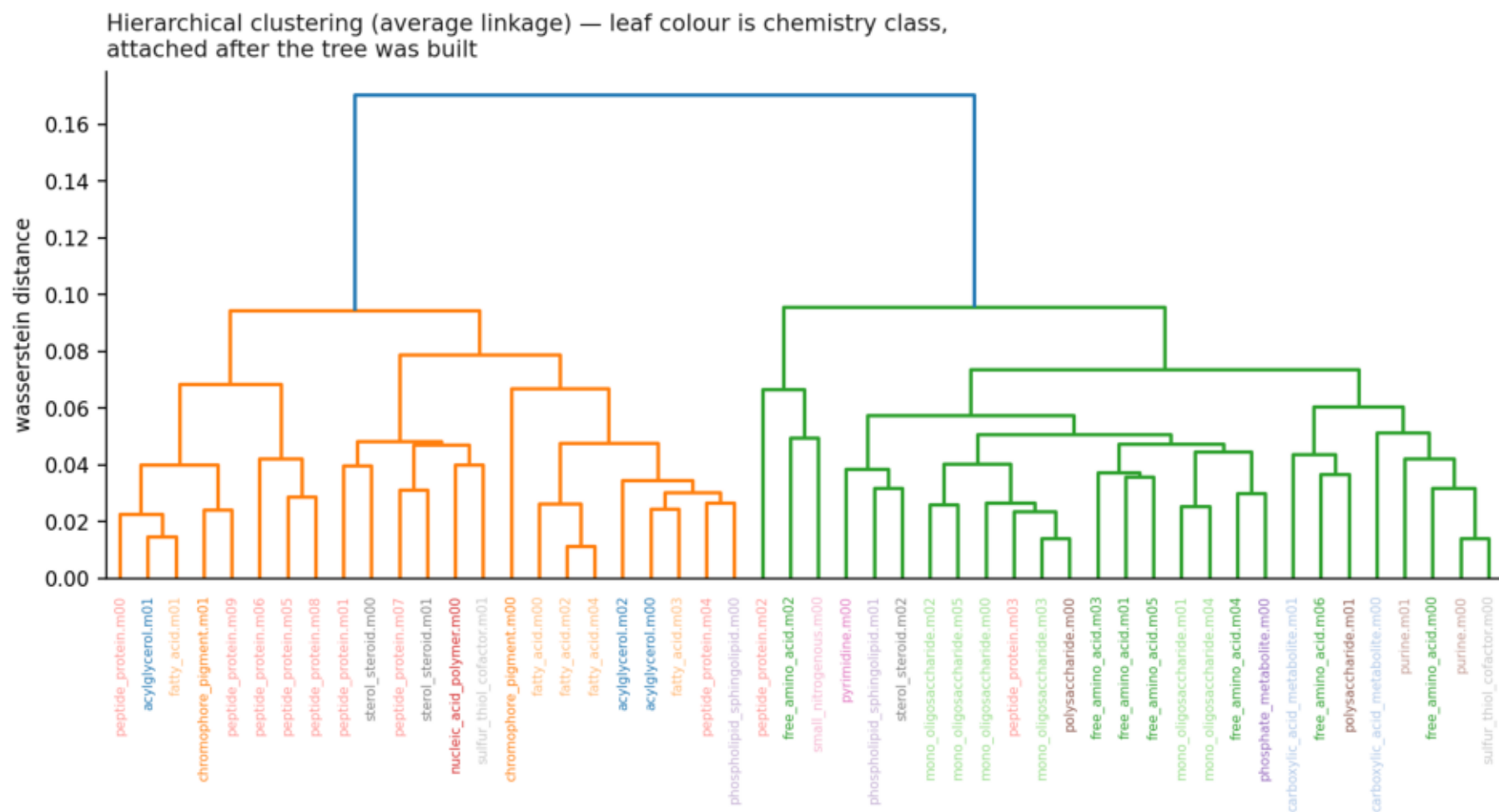
UMAP stability sweep over $n_neighbors \times min_dist \times seed$. Neighbourhoods survive the parameters (Jaccard 0.60–0.65 against the high-dimensional geometry); the global arrangement does not (Procrustes disparity 0.45).

Figure 07



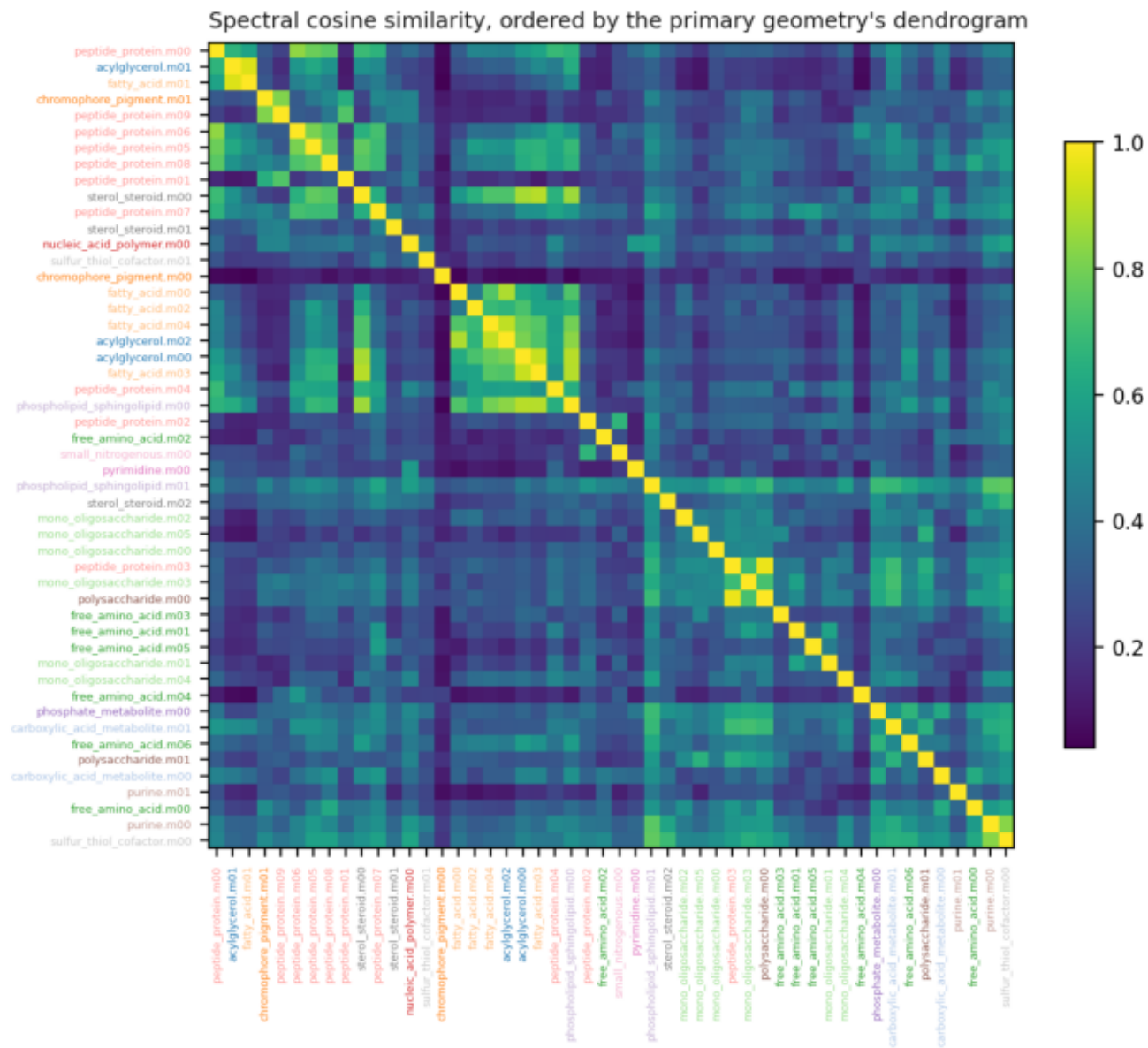
Diffusion map and its eigenvalue spectrum. The absence of a sharp gap (1.000, 0.186, 0.150, 0.119) is the first quantitative evidence against discrete clusters.

Figure 08



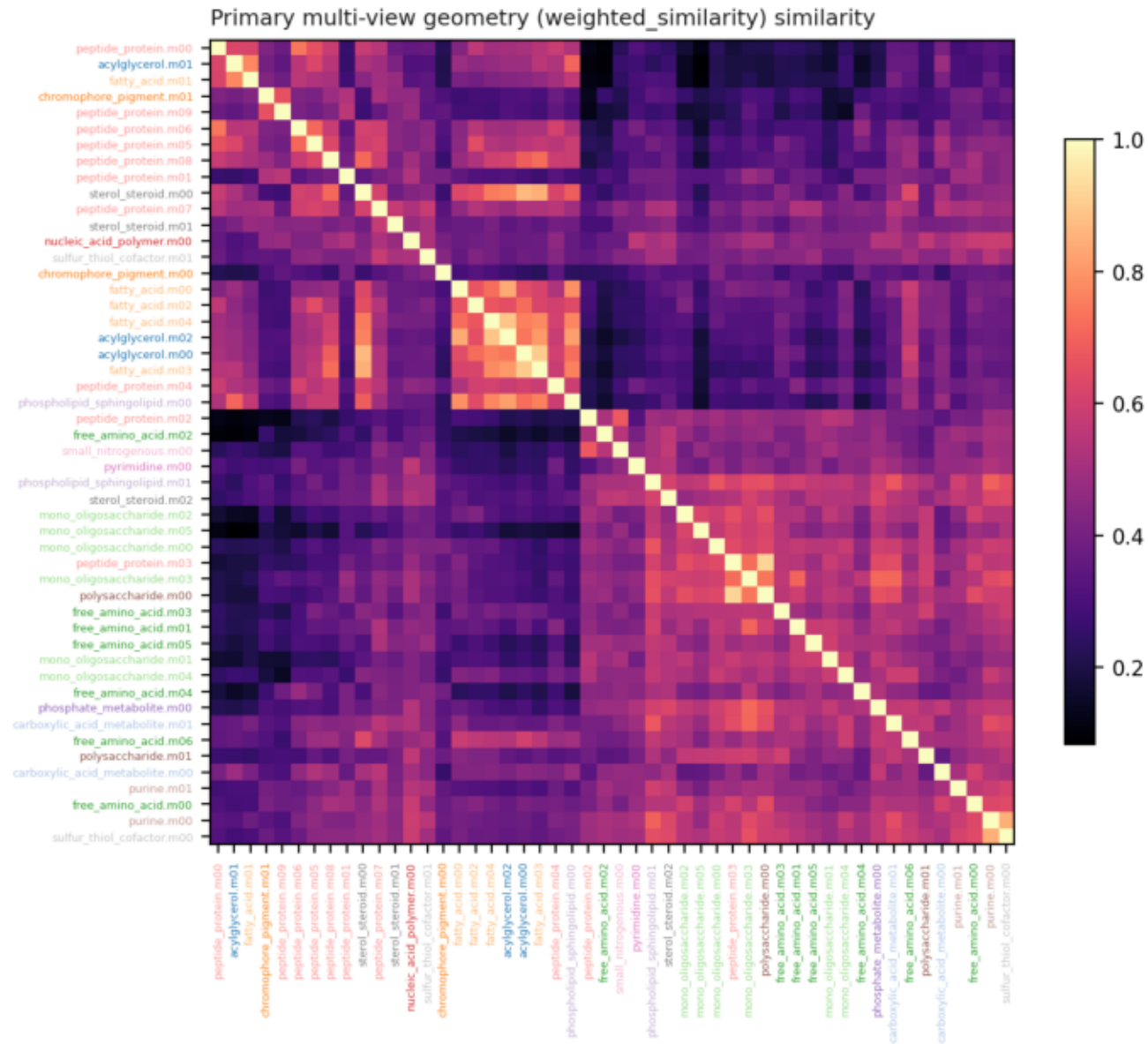
Hierarchical clustering. Leaf colour is chemistry class, attached after the tree was built.

Figure 09



Spectral cosine similarity, ordered by the primary geometry's dendrogram. Cosine is the worst metric on background separation (0.008) — it is dominated by the shared broad envelope.

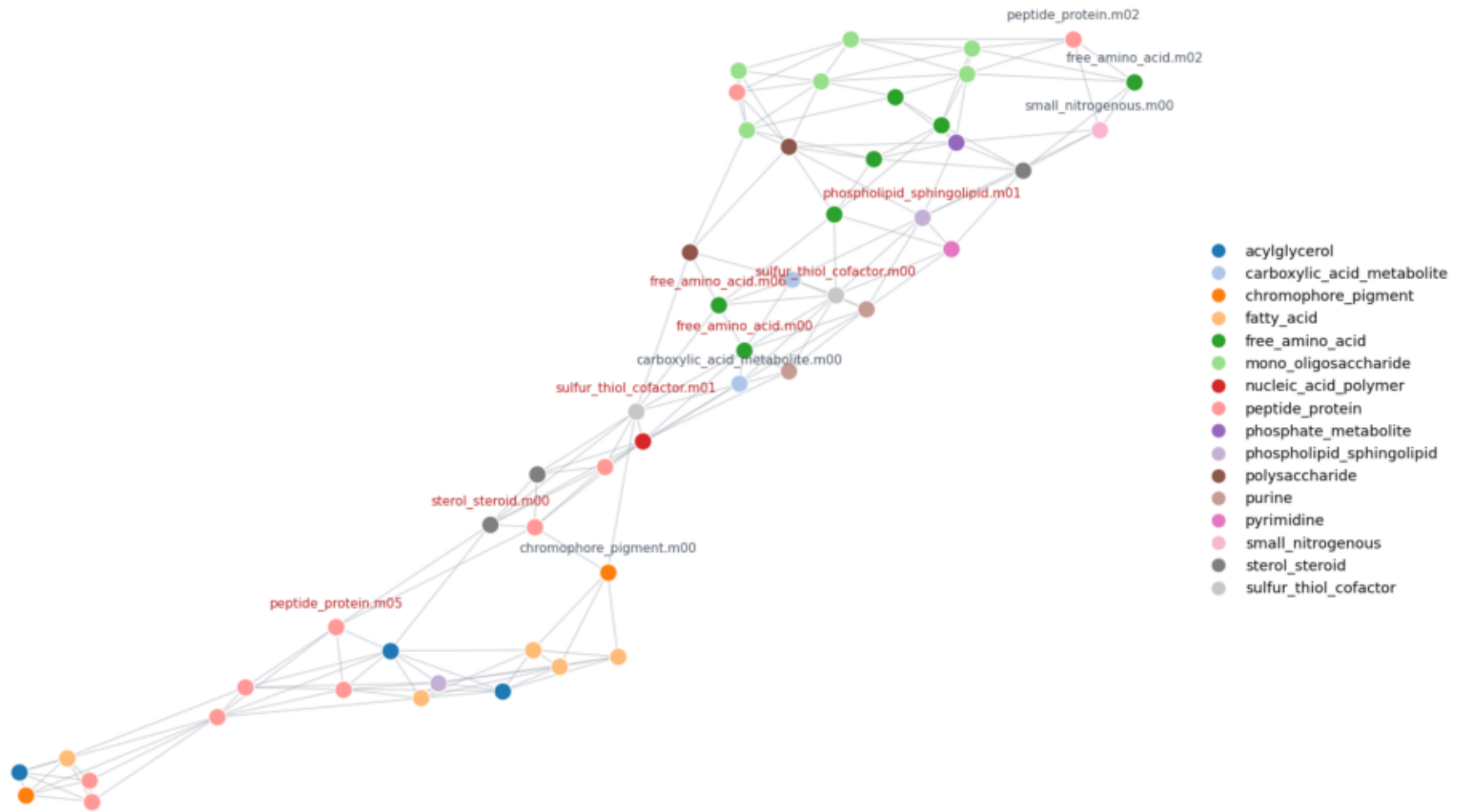
Figure 10



The primary multi-view geometry (weighted similarity fusion).

Figure 11

k-nearest-neighbour graph (k = 5). Labelled: bridge motifs (red) and isolated motifs (grey).



k-nearest-neighbour graph (k = 5). Modularity 0.436 against a degree-preserving null of 0.070 ± 0.003 . Labelled: bridges in red, isolates in grey.

Figure 12

Force-directed motif graph (Kamada-Kawai)



Force-directed layout of the same graph (Kamada-Kawai).

Figure 13

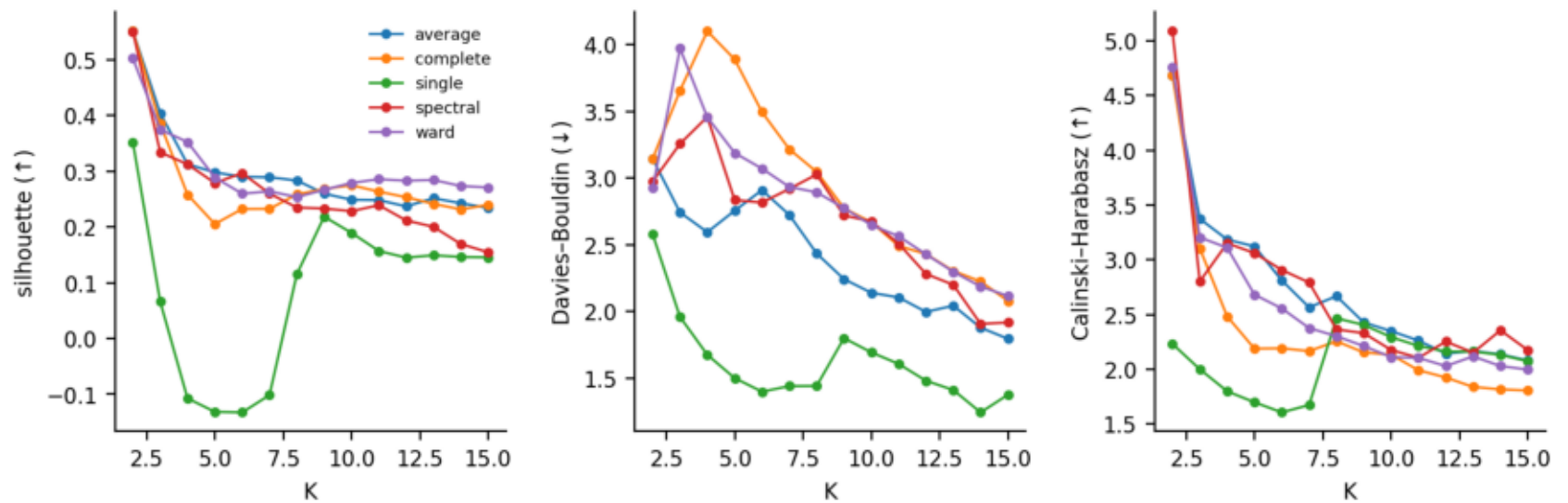
Minimum spanning tree — the backbone of motif space



Minimum spanning tree — the backbone of motif space.

Figure 14

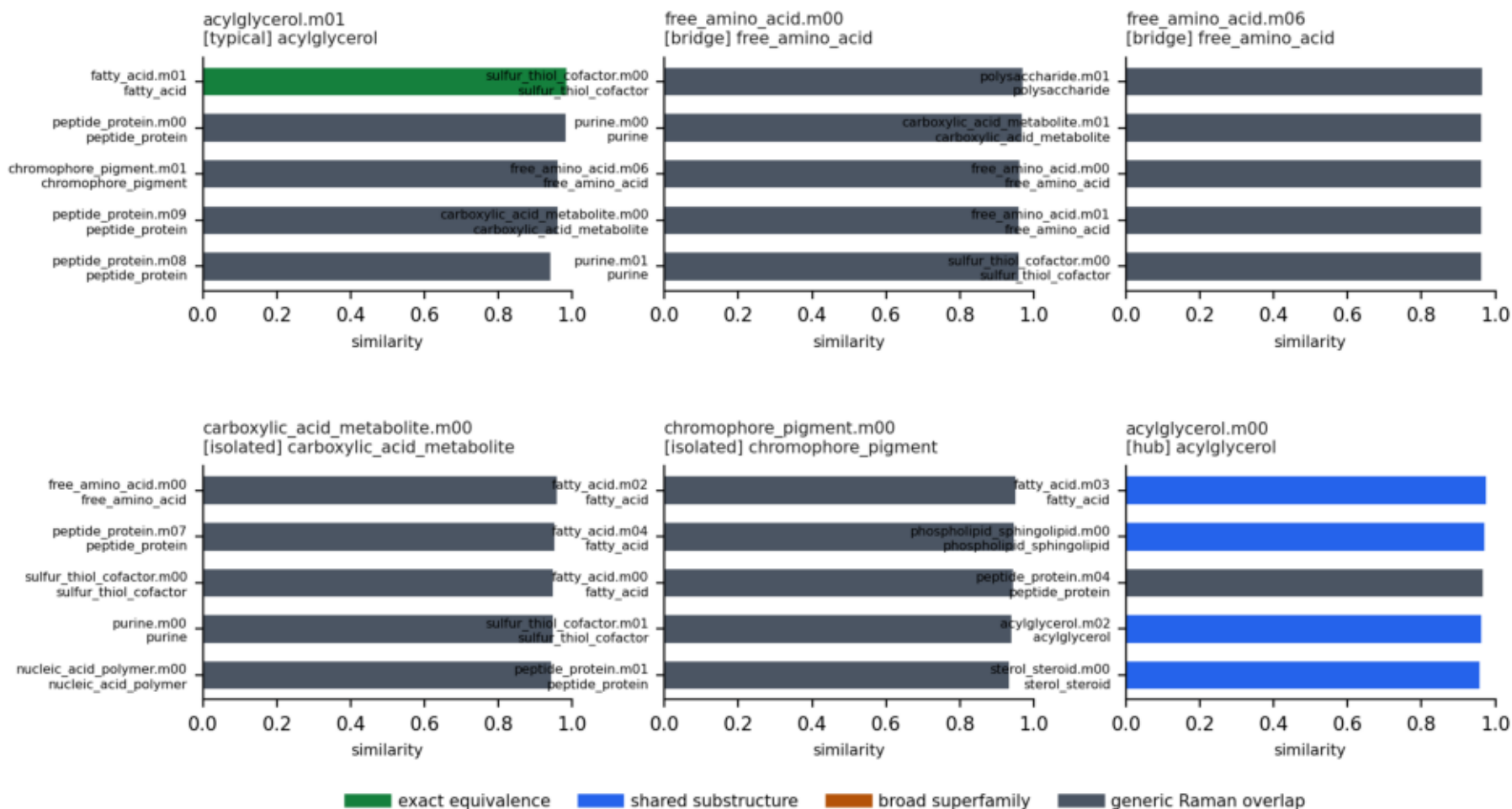
Cluster quality across K and linkage — silhouette and Calinski-Harabasz both pick K = 2 (a hydrophobic/polar split, bootstrap ARI 0.879); Davies-Bouldin runs to the largest K, which is what it does when there is no further structure. One defensible cut, at the top.



Cluster quality across K and linkage. Silhouette (0.551) and Calinski-Harabasz agree on K = 2 under four of five linkages, with bootstrap ARI 0.879; Davies-Bouldin runs monotonically to the largest K, its behaviour when no further structure exists. One defensible cut, at the top.

Figure 15

Nearest-neighbour cards — five neighbours per motif, each classified by what kind of relationship it is



Nearest-neighbour cards. 188 of 250 links (75%) are indistinguishable from the null — the same lesson Phase 02 learned at the edge level, one level up.

Figure 16

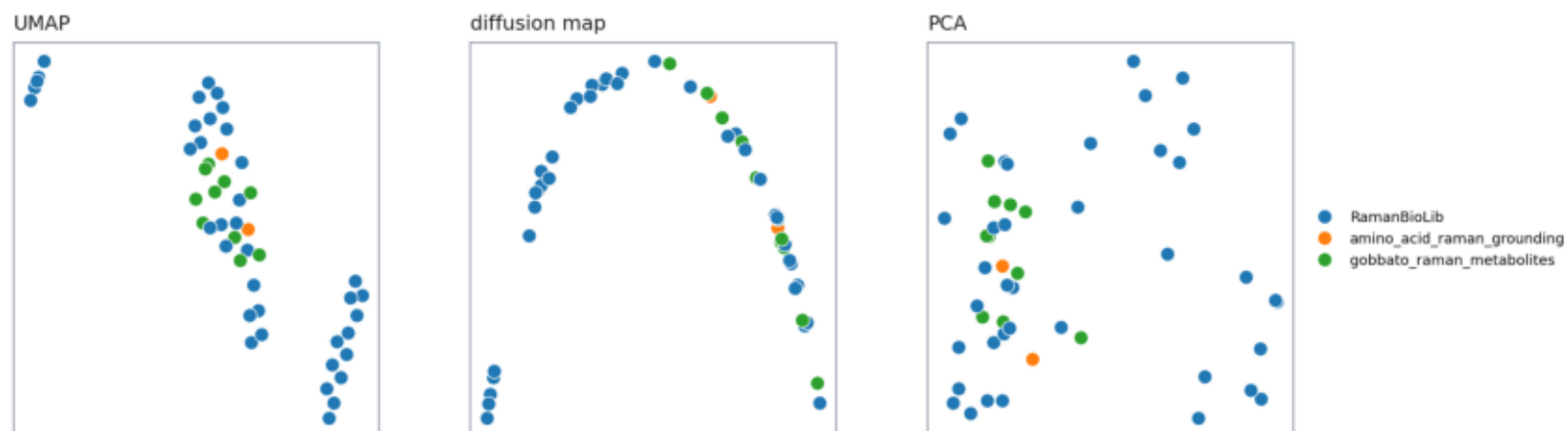
Geometry coloured by chemistry class — PERMANOVA $F = 3.65$, $p = 0.001$, $R^2 = 0.617$; kNN 0.240 vs chance 0.102



Geometry coloured by chemistry class. PERMANOVA $R^2 = 0.617$, $p = 0.001$.

Figure 17

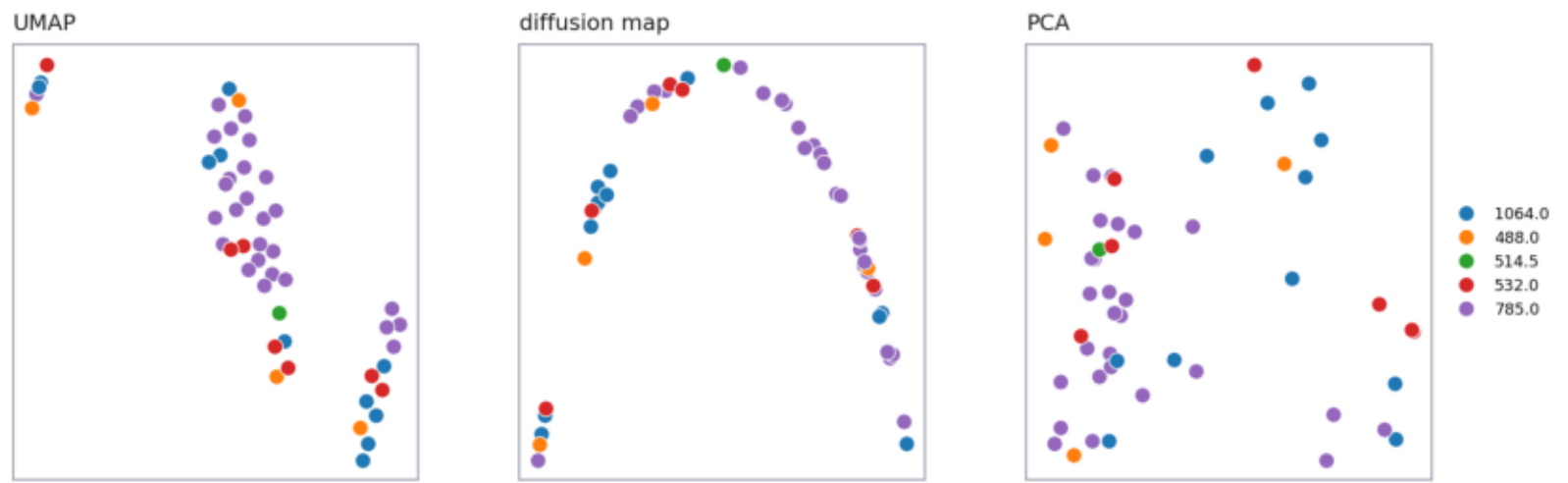
Geometry coloured by SOURCE — PERMANOVA $F = 3.53$, $p = 0.022$, $R^2 = 0.130$. Significant but far weaker than chemistry: a caution, not a veto.



Geometry coloured by source. $R^2 = 0.130$, $p = 0.022$ — significant but $4.7\times$ weaker than chemistry. A caution, not a veto. 8 of 50 motifs are single-source and therefore untestable.

Figure 18

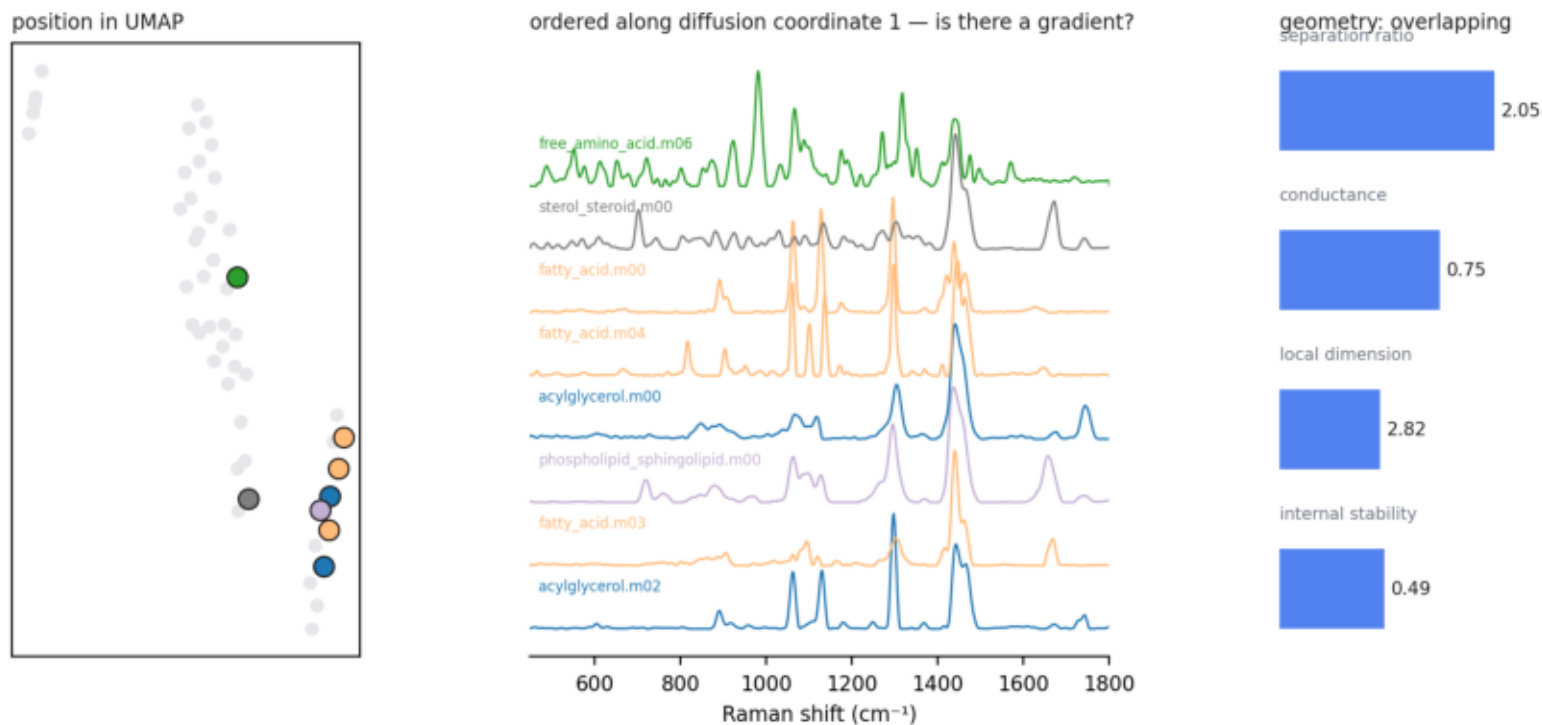
Geometry coloured by EXCITATION — PERMANOVA $F = 2.43$, $p = 0.044$, $R^2 = 0.178$



Geometry coloured by excitation. $R^2 = 0.178$, $p = 0.044$.

Figure 19

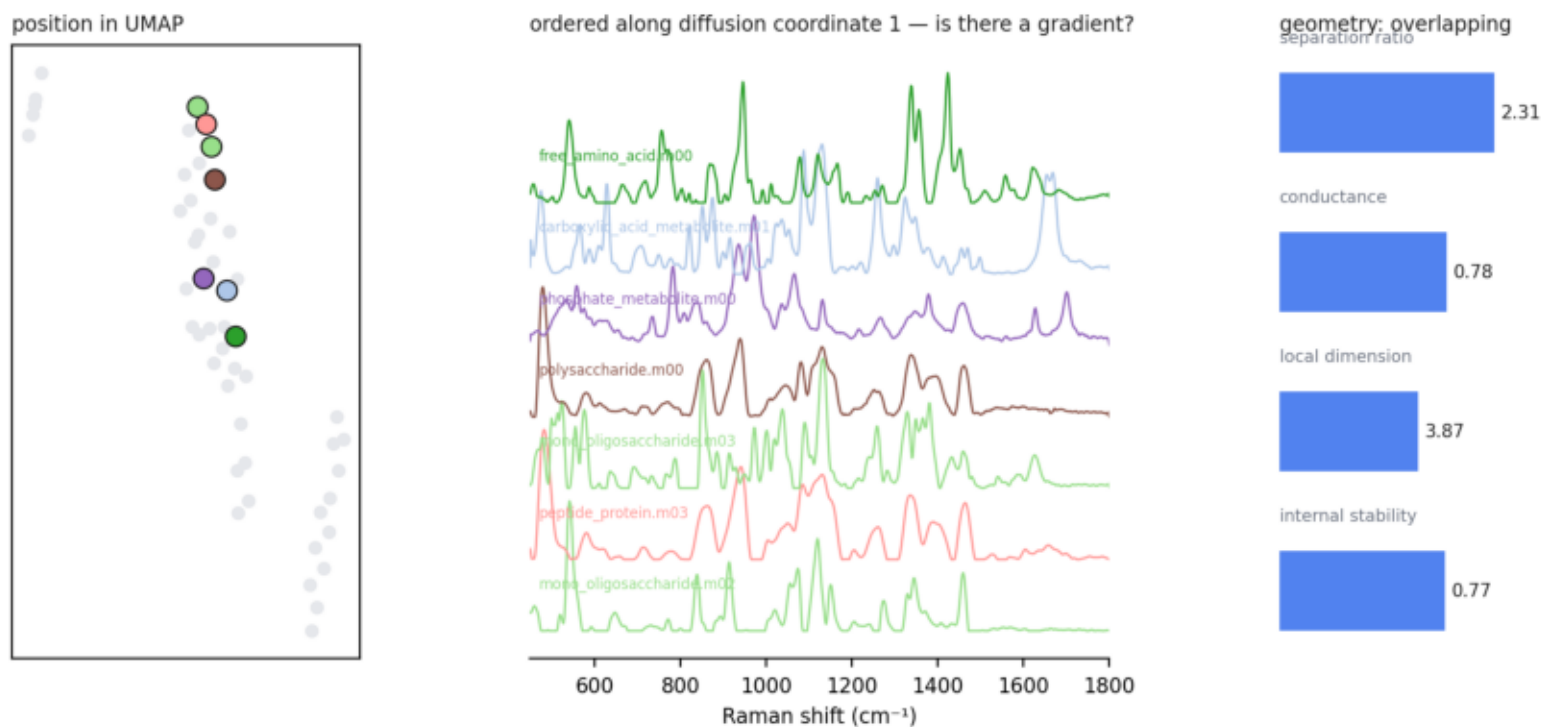
Lipid neighbourhood (Phase 02 proposal00) — rejected as a merge, examined here as geometry



Lipid neighbourhood (Phase 02 proposal00). Separation ratio 2.05, local dimension 2.82 — the most low-dimensional group in the corpus. Ordered along diffusion coordinate 1 it runs amino acid → sterol → free fatty acids → acylglycerols: an aliphatic chain-order gradient, with the sterol adjacent but outside.

Figure 20

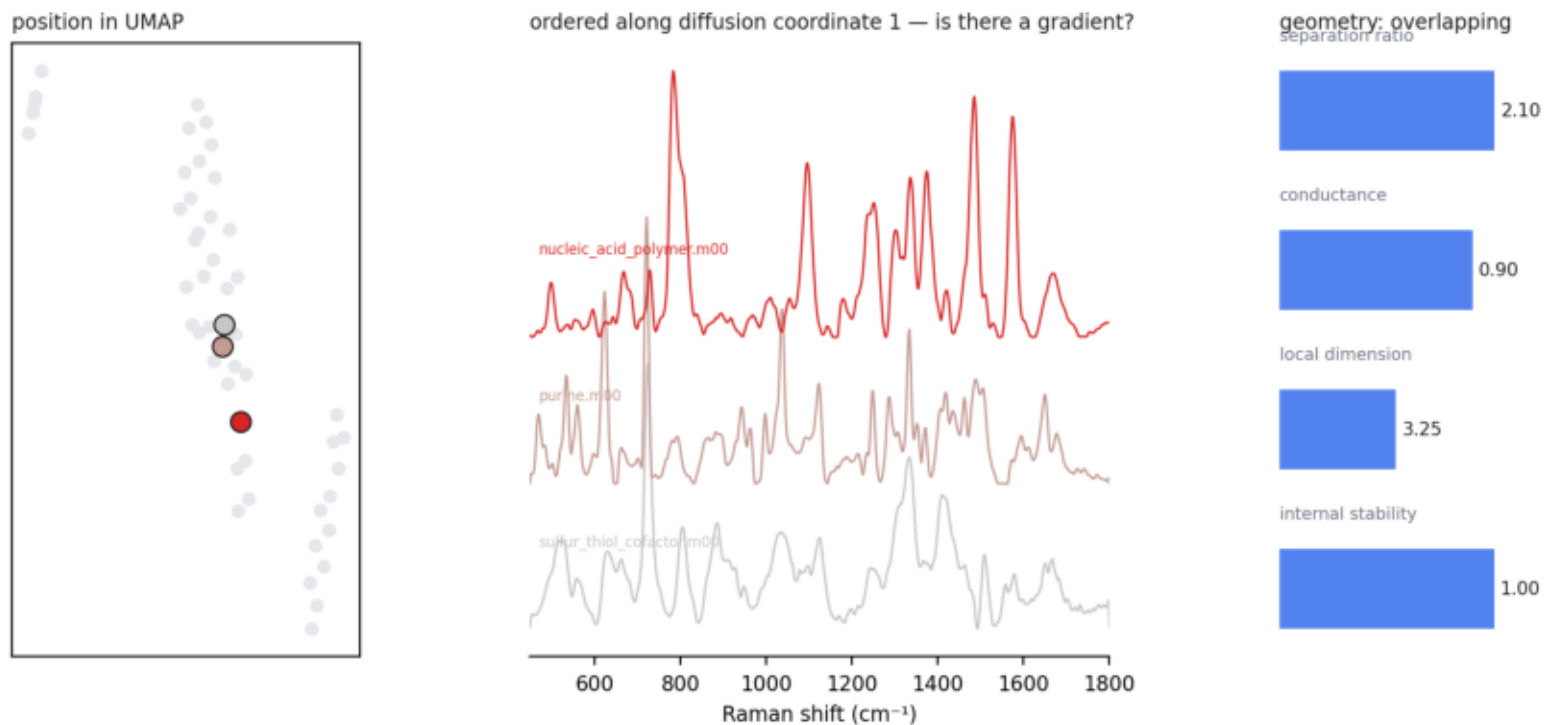
Polar skeletal neighbourhood (proposal03) — protein, saccharide, carboxylate, phosphate



Polar skeletal neighbourhood (proposal03). Separation ratio 2.31, the strongest of the three. Shared bands at $1074/1120 \text{ cm}^{-1}$ confirm that protein and polysaccharide are close through skeletal C-O/C-C overlap, not glycoprotein biology.

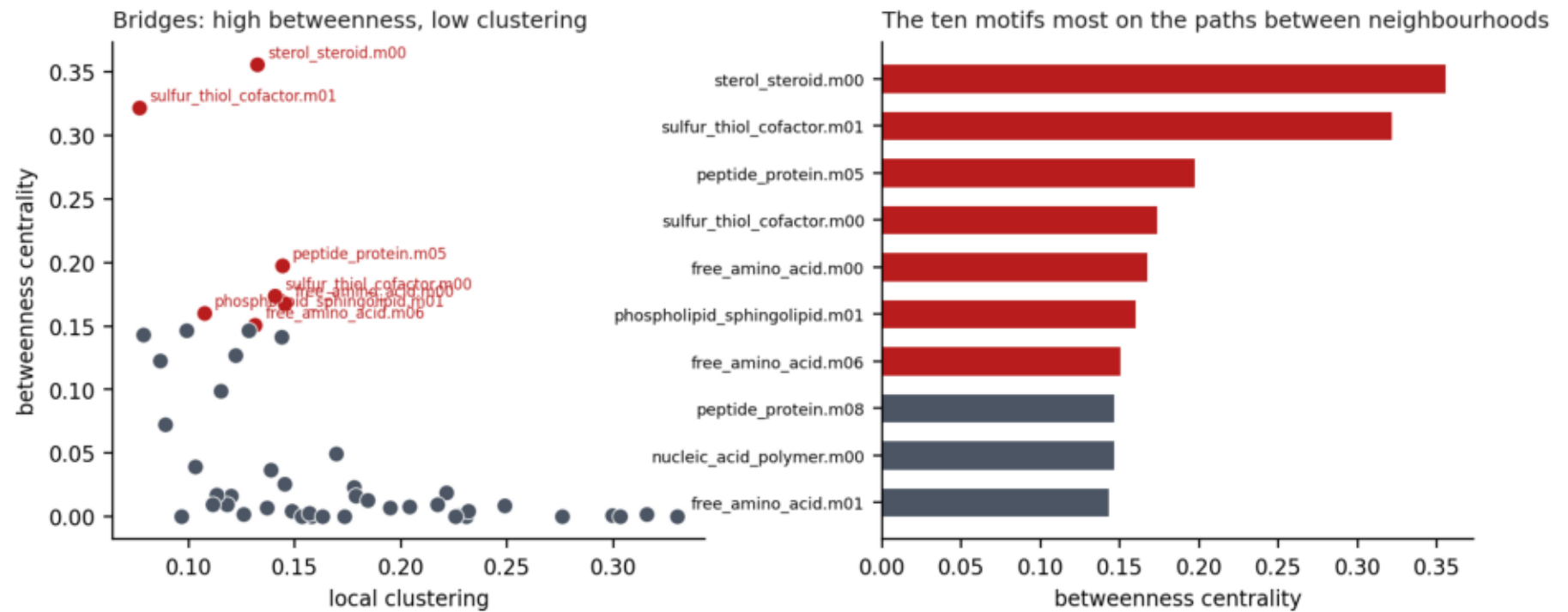
Figure 21

Heterocyclic / ring-system neighbourhood (proposal16) — nucleic acid, purine, thiol cofactor



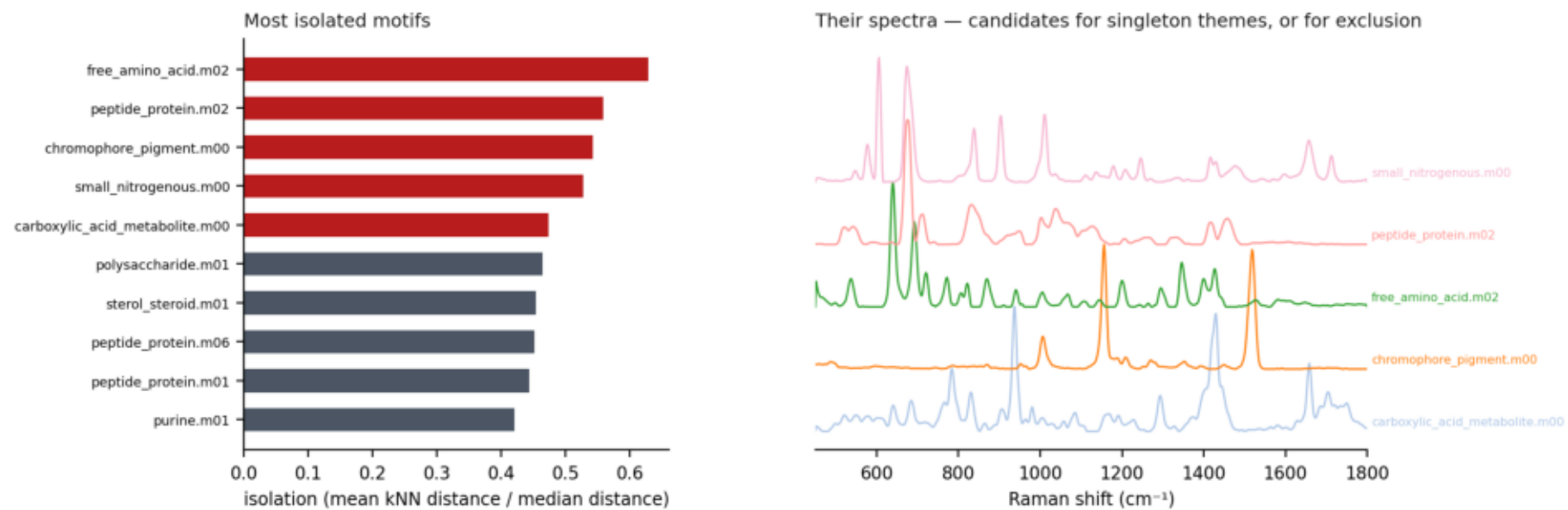
Heterocyclic ring-system neighbourhood (proposal16). Internal stability 1.000. The cofactor motifs sit near purine because coenzyme A and acetyl-CoA contain adenine — which is why merging the group destroyed reconstruction.

Figure 22



Bridge motifs: high betweenness, low local clustering. Seven motifs sit on the paths between neighbourhoods without belonging to one. A hard theme assignment would misplace each of them.

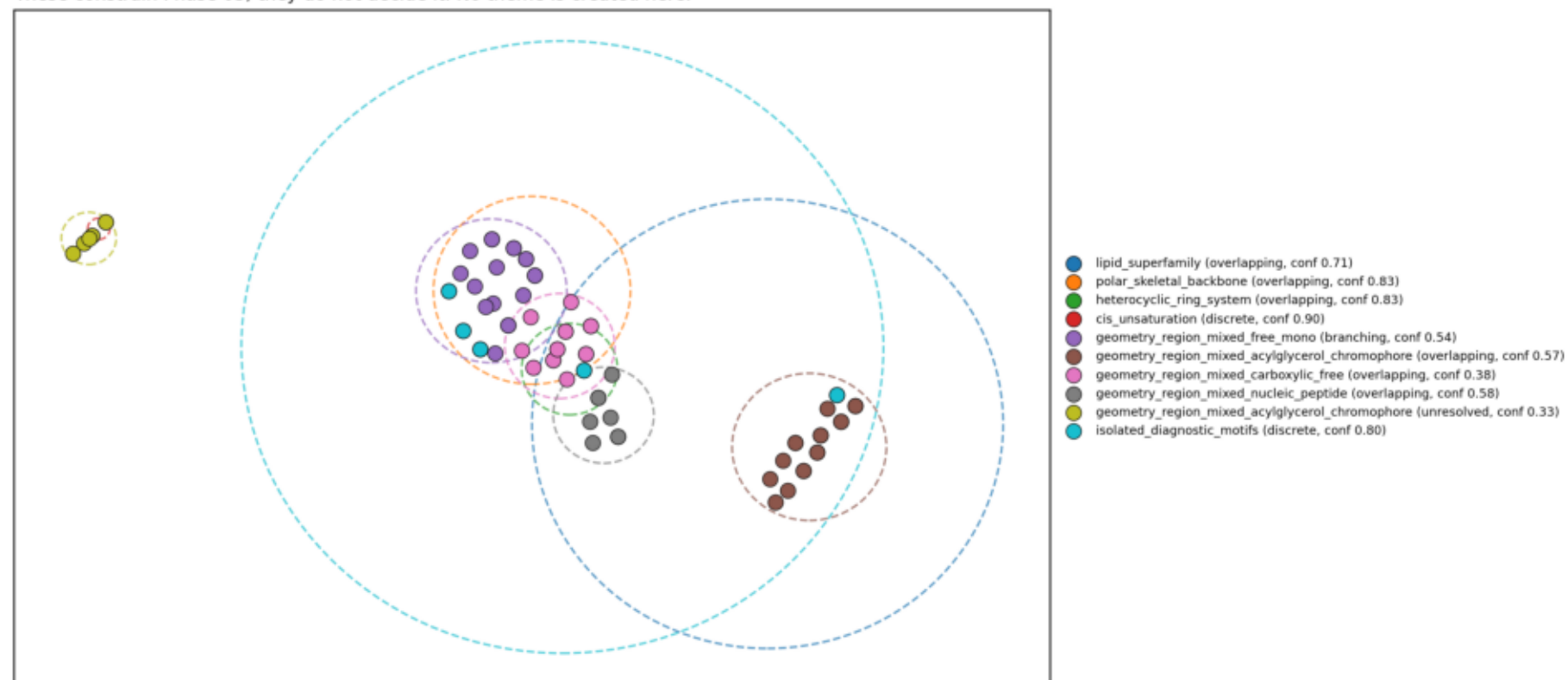
Figure 23



Isolated motifs. Five have no close neighbour; chromophore_pigment.m00 is the clearest case — a conjugated carotenoid C=C system with no analogue in this corpus. Candidates for singleton themes or for exclusion.

Figure 24

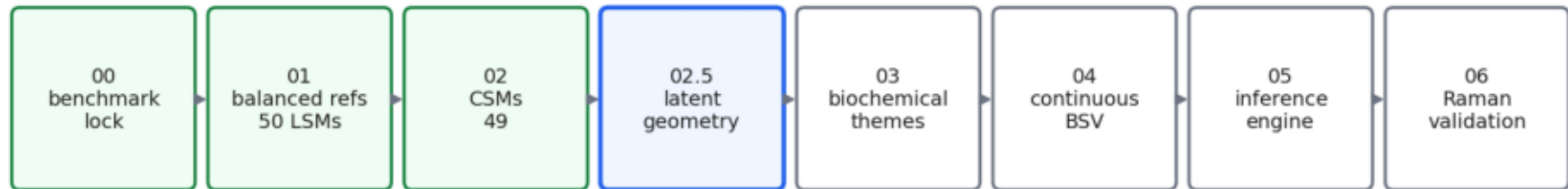
Provisional Phase 03 priors on the primary geometry.
These constrain Phase 03; they do not decide it. No theme is created here.



The ten provisional Phase 03 priors on the primary geometry. These constrain Phase 03; they do not decide it.

Figure 25

GAIRA V7 architecture after Phase 02.5



Phase 02.5 is an inserted ANALYSIS phase. It produces geometry and priors, not fitted objects;
Phase 03 consumes the priors as constraints and remains free to reject them.

primary metric: wasserstein · primary geometry: weighted_similarity · 10 provisional priors